# Probability and Statistics

## Chapter 9. One- and Two-Sample Estimation Problems

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# Chapter Overview

# 9.1 Introduction

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In previous chapters, we studied sampling properties of the sample mean and sample variance. These ideas provide the foundation for drawing conclusions about population parameters from experimental data.

- The **Central Limit Theorem** describes the sampling distribution of $\bar{X}$.
- The sampling distribution of $\bar{X}$ involves the population mean $\mu$.
- The sampling distribution of $S^2$ is used when making conclusions about $\sigma^2$.

이 장의 목적은 표본에서 얻은 정보를 이용하여 **population parameter**를 추정하는 방법을 배우는 것이다.

# Purpose of Chapter 9

This chapter begins the formal discussion of **statistical inference**. The focus is on estimation procedures involving one sample and two samples.

Main topics:

- point estimation
- interval estimation
- confidence intervals for means, proportions, and variances
- prediction intervals and tolerance limits
- maximum likelihood estimation

표본평균, 표본분산, 표본비율은 단순한 계산값이 아니라 모집단 모수를 추정하기 위한 statistic이다.

# 9.2 Statistical Inference

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**Statistical inference** consists of methods by which we make generalizations about a population from sample data.

Two broad approaches:

- **Classical method**: inference is based on random sample information.
- **Bayesian method**: inference uses prior information together with sample information.

이 장에서는 주로 classical methods를 사용하여 mean, proportion, variance를 추정한다.

# Estimation and Tests of Hypotheses

Statistical inference may be divided into two major areas.

- **Estimation**: estimating an unknown population parameter.
- **Tests of hypotheses**: making a decision about a stated claim.

- Estimation example: estimate the true proportion of voters supporting a candidate.
- Hypothesis testing example: decide whether brand A floor wax is more scuff-resistant than brand B.

estimation은 모수의 값을 추정하는 것이고, hypothesis testing은 미리 세운 주장에 대해 판단하는 것이다.

# 9.3 Classical Methods of Estimation

# Point Estimate and Estimator

A **point estimate** of a population parameter $\theta$ is a single value $\hat{\theta}$ of a statistic $\hat{\Theta}$.

Examples:

- $\bar{x}$ is a point estimate of $\mu$.
- $\hat{p} = x/n$ is a point estimate of the binomial parameter $p$.

Estimator는 random sample의 함수이고, estimate는 실제 표본값을 대입하여 얻은 숫자이다.

# Unbiased Estimator

A statistic $\hat{\Theta}$ is an **unbiased estimator** of the parameter $\theta$ if

$$E(\hat{\Theta}) = \theta.$$

- Unbiasedness means that the estimator is correct on average.
- It does not mean that every individual estimate equals the true parameter.

unbiased estimator는 반복 표본추출을 했을 때 평균적으로 true parameter를 맞추는 estimator이다.

# Example 9.1

표본분산 $S^2$가 population variance $\sigma^2$의 unbiased estimator임을 보이시오.

- 즉, 보여야 할 것은

$$E(S^2) = \sigma^2$$

  이다.

- 여기서

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

# Example 9.1: Solution (1)

Section 8.5에서 다음 항등식을 사용하였다.

$$\sum_{i=1}^{n}(X_i - \bar{X})^2 = \sum_{i=1}^{n}(X_i - \mu)^2 - n(\bar{X} - \mu)^2.$$

따라서

$$E(S^2) = E\left[\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2\right]$$

$$= \frac{1}{n-1}\left[\sum_{i=1}^{n}E(X_i - \mu)^2 - nE(\bar{X} - \mu)^2\right].$$

# Example 9.1: Solution (2)

각 관측값은 같은 population variance를 가지므로

$$E(X_i - \mu)^2 = \sigma^2, \qquad i = 1, 2, \ldots, n.$$

또한 sample mean의 variance는

$$E(\bar{X} - \mu)^2 = \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

따라서

$$E(S^2) = \frac{1}{n-1}\left(n\sigma^2 - n\frac{\sigma^2}{n}\right) = \frac{1}{n-1}(n\sigma^2 - \sigma^2) = \sigma^2.$$

따라서 $S^2$는 $\sigma^2$의 unbiased estimator이다. 이것이 표본분산에서 $n$이 아니라 $n-1$로 나누는 이유이다.

# Variance and Efficiency of an Estimator

If two estimators are both unbiased, we prefer the one with smaller variance.

If $\hat{\Theta}_1$ and $\hat{\Theta}_2$ are unbiased estimators of $\theta$ and

$$\mathrm{Var}(\hat{\Theta}_1) < \mathrm{Var}(\hat{\Theta}_2),$$

then $\hat{\Theta}_1$ is more efficient than $\hat{\Theta}_2$.

좋은 estimator는 평균적으로 맞을 뿐 아니라, 표본이 바뀌어도 값이 크게 흔들리지 않아야 한다.

# Interval Estimation

A point estimate gives one number, but it usually does not equal the true parameter exactly. Therefore, it is often preferable to give an interval.

An interval estimate of $\theta$ has the form

$$\hat{\theta}_L < \theta < \hat{\theta}_U.$$

- The endpoints depend on the observed statistic.
- The length of the interval reflects estimation accuracy.

# Confidence Interval Interpretation

If

$$P(\hat{\Theta}_L < \theta < \hat{\Theta}_U) = 1 - \alpha,$$

then the computed interval is called a $100(1 - \alpha)\%$ **confidence interval**.

Terminology:

- $1 - \alpha$: confidence coefficient
- $\hat{\theta}_L$: lower confidence limit
- $\hat{\theta}_U$: upper confidence limit

95% confidence interval은 “모수가 이 구간에 있을 확률이 95%”라는 뜻이 아니라, 같은 절차를 반복하면 약 95%의 구간이 true parameter를 포함한다는 뜻이다.

# 9.4 Single Sample: Estimating the Mean

# Estimating a Single Mean

The sample mean $\bar{X}$ is used as a point estimator of the population mean $\mu$.

$$E(\bar{X}) = \mu, \qquad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

- A larger sample size reduces the variance of $\bar{X}$.
- Hence $\bar{x}$ is likely to be closer to $\mu$ when $n$ is large.

# Confidence Interval on $\mu$: $\sigma^2$ Known

If the population variance $\sigma^2$ is known and the sample is from a normal population, or $n$ is sufficiently large, then

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

has approximately the standard normal distribution.

A $100(1-\alpha)\%$ confidence interval for $\mu$ is

$$\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

# Example 9.2

한 강의 위치한 36개 지점에서 zinc concentration을 측정한 결과 sample mean이 2.6 grams per milliliter였다. Population standard deviation은 0.3 gram per milliliter라고 가정한다.

1. Mean zinc concentration $\mu$에 대한 95% confidence interval을 구하시오.
2. Mean zinc concentration $\mu$에 대한 99% confidence interval을 구하시오.

# Example 9.2: Solution

Given

$$\bar{x} = 2.6, \qquad \sigma = 0.3, \qquad n = 36.$$

For 95%, $z_{0.025} = 1.96$:

$$2.6 - (1.96)\frac{0.3}{\sqrt{36}} < \mu < 2.6 + (1.96)\frac{0.3}{\sqrt{36}}$$
$$2.50 < \mu < 2.70.$$

For 99%, $z_{0.005} = 2.575$:

$$2.47 < \mu < 2.73.$$

신뢰수준이 높아지면 더 넓은 interval이 필요하다.

# Error Bound for Estimating $\mu$

If $\bar{x}$ is used as an estimate of $\mu$, the estimation error is

$$|\bar{x} - \mu|.$$

We can be $100(1-\alpha)\%$ confident that the error will not exceed

$$z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

confidence interval의 반폭은 estimation error의 upper bound로 해석할 수 있다.

# Sample Size for Estimating $\mu$

If we want the error to be no larger than a specified amount $e$, choose $n$ so that

$$z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = e.$$

The required sample size is

$$n = \left(\frac{z_{\alpha/2}\sigma}{e}\right)^2.$$

- If $n$ is not an integer, round up.
- Rounding down may reduce the desired degree of confidence.

# Example 9.3

Example 9.2에서 $\mu$의 estimate가 true mean과 0.05 미만의 오차를 갖도록 95% confident 하고 싶다. 필요한 sample size는 얼마인가?

Given $\sigma = 0.3$, $e = 0.05$, and $z_{0.025} = 1.96$:

$$n = \left(\frac{(1.96)(0.3)}{0.05}\right)^2 = 138.3.$$

따라서 sample size는 올림하여

$$n = 139$$

가 필요하다.

# One-Sided Confidence Bounds on $\mu$

Sometimes only one direction matters.

- Tensile strength: lower bound may be most important.
- Pollution concentration: upper bound may be most important.

If $\sigma^2$ is known, one-sided $100(1-\alpha)\%$ bounds are

$$\text{upper bound: } \bar{x} + z_\alpha \frac{\sigma}{\sqrt{n}}, \qquad \text{lower bound: } \bar{x} - z_\alpha \frac{\sigma}{\sqrt{n}}.$$

# Example 9.4

25명의 subject를 선택하여 특정 자극에 대한 reaction time을 측정하였다. 과거 경험상 variance는 4 sec$^2$이고, distribution은 approximately normal이다. Sample mean은 6.2 seconds이다. Mean reaction time의 upper 95% bound를 구하시오.

Given $\bar{x} = 6.2$, $\sigma = 2$, $n = 25$, and $z_{0.05} = 1.645$:

$$\bar{x} + z_\alpha \frac{\sigma}{\sqrt{n}} = 6.2 + (1.645)\frac{2}{5} = 6.858.$$

따라서 mean reaction time이 6.858 seconds보다 작다고 95% confident할 수 있다.

# Confidence Interval on $\mu$: $\sigma^2$ Unknown

When $\sigma$ is unknown and the sample comes from a normal population, use

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}.$$

Then $T$ follows a t-distribution with

$$v = n - 1$$

degrees of freedom.

A $100(1-\alpha)\%$ confidence interval is

$$\bar{x} - t_{\alpha/2}\frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2}\frac{s}{\sqrt{n}}.$$

# Example 9.5

7개의 비슷한 sulfuric acid container의 내용량이 다음과 같다.

$$9.8, 10.2, 10.4, 9.8, 10.0, 10.2, 9.6$$

Approximately normal distribution을 가정할 때, 모든 container의 mean content에 대한 95% confidence interval을 구하시오.

# Example 9.5: Solution

For the data,

$$\bar{x} = 10.0, \qquad s = 0.283, \qquad n = 7.$$

Degrees of freedom:

$$v = n - 1 = 6.$$

From the t-table,

$$t_{0.025} = 2.447.$$

$$10.0 - (2.447)\frac{0.283}{\sqrt{7}} < \mu < 10.0 + (2.447)\frac{0.283}{\sqrt{7}}$$

$$9.74 < \mu < 10.26.$$

# Large-Sample Confidence Interval

When $\sigma$ is unknown and $n \geq 30$, it is common to replace $\sigma$ by $s$ and use the normal approximation.

Large-sample confidence interval:

$$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}.$$

$n$이 충분히 크고 distribution이 심하게 skewed하지 않으면 $s \approx \sigma$로 보고 normal approximation을 사용할 수 있다.

# Example 9.6

Texas high school senior 500명의 SAT mathematics score에서 sample mean은 501, sample standard deviation은 112였다. Mean SAT mathematics score에 대한 99% confidence interval을 구하시오.

Since $n = 500$ is large, use normal approximation. For 99%, $z_{0.005} = 2.575$:

$$501 \pm (2.575)\frac{112}{\sqrt{500}} = 501 \pm 12.9.$$

따라서 99% confidence interval은

$$488.1 < \mu < 513.9.$$

# Single Mean Estimation

- $\sigma$ known: $z$-interval을 사용한다.
- $\sigma$ unknown and normal population: $t$-interval을 사용한다.
- $n \geq 30$이면 $s$를 사용한 large-sample $z$-interval을 근사적으로 사용할 수 있다.
- Confidence interval의 폭은 $n$, variability, confidence level에 의해 결정된다.

# 9.5 Standard Error of a Point Estimate

# 9.5 Standard Error of a Point Estimate

The **standard error** of an estimator is the standard deviation of its sampling distribution.

For the sample mean,

$$\mathrm{s.e.}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

If $\sigma$ is unknown, use the estimated standard error

$$\mathrm{s.e.}(\bar{x}) = \frac{s}{\sqrt{n}}.$$

standard error는 point estimate가 표본마다 얼마나 흔들리는지를 나타내는 양이다.

# Confidence Limits Using Standard Error

The confidence interval can be written using standard error.

When $\sigma$ is known:

$$\bar{x} \pm z_{\alpha/2}\mathrm{s.e.}(\bar{x}).$$

When $\sigma$ is unknown and normality is assumed:

$$\bar{x} \pm t_{\alpha/2}\mathrm{s.e.}(\bar{x}).$$

Interval width depends directly on the standard error. A smaller standard error produces a narrower confidence interval.

# 9.6 Prediction Intervals

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Confidence intervals estimate a population mean. Sometimes we want to predict a future observation.

A **prediction interval** gives a plausible range for a single future value $x_0$.

- It includes uncertainty from estimating $\mu$.
- It also includes the variability of a future single observation.

confidence interval은 mean에 대한 구간이고, prediction interval은 다음 관측값 하나에 대한 구간이다.

# Prediction Interval: $\sigma^2$ Known

For a normal distribution with unknown mean $\mu$ and known variance $\sigma^2$,

$$Z = \frac{x_0 - \bar{x}}{\sigma\sqrt{1 + 1/n}}$$

follows the standard normal distribution.

A $100(1-\alpha)\%$ prediction interval for a future observation $x_0$ is

$$\bar{x} - z_{\alpha/2}\sigma\sqrt{1 + \frac{1}{n}} < x_0 < \bar{x} + z_{\alpha/2}\sigma\sqrt{1 + \frac{1}{n}}.$$

# Example 9.7

First Citizens Bank는 최근 50개의 mortgage loan sample을 조사했고, average loan amount는 $257,300이었다. Population standard deviation은 $25,000이라고 가정한다.

다음 고객의 loan amount에 대한 95% prediction interval을 구하시오.

# Example 9.7: Solution

Given

$$\bar{x} = 257300, \qquad \sigma = 25000, \qquad n = 50, \qquad z_{0.025} = 1.96.$$

Prediction interval:

$$257300 \pm (1.96)(25000)\sqrt{1 + \frac{1}{50}}.$$

$$207812.43 < x_0 < 306787.57.$$

이 구간은 평균 loan amount가 아니라, 다음 고객 한 명의 loan amount를 예측하는 구간이다.

# Prediction Interval: $\sigma^2$ Unknown

If $\sigma^2$ is unknown and the population is normal, replace $\sigma$ by $s$ and use the t-distribution.

A $100(1-\alpha)\%$ prediction interval for a future observation $x_0$ is

$$\bar{x} - t_{\alpha/2} s\sqrt{1+\frac{1}{n}} < x_0 < \bar{x} + t_{\alpha/2} s\sqrt{1+\frac{1}{n}}.$$

One-sided bounds:

$$\bar{x} + t_{\alpha} s\sqrt{1+\frac{1}{n}}, \qquad \bar{x} - t_{\alpha} s\sqrt{1+\frac{1}{n}}.$$

# Example 9.8

A meat inspector가 95% lean beef 30개를 무작위로 선택하였다. Sample mean은 96.2%, sample standard deviation은 0.8%였다. Normality를 가정하고 새 pack의 leanness에 대한 99% prediction interval을 구하시오.

For $v = 29$, $t_{0.005} = 2.756$:

$$96.2 \pm (2.756)(0.8)\sqrt{1 + \frac{1}{30}}.$$

$$93.96 < x_0 < 98.44.$$

# Prediction Limits and Outlier Detection

Prediction intervals can be used to detect outliers.

An observation is treated as an outlier if it falls outside the prediction interval computed without including the questionable observation.

- If a new beef pack in Example 9.8 has leanness outside $(93.96, 98.44)$, it can be regarded as an outlier.
- The logic is based on whether the new observation is plausible under the same population model.

# 9.7 Tolerance Limits

# 9.7 Tolerance Limits

A tolerance interval is used when we want to cover a specified proportion of the population.

For a normal distribution with unknown $\mu$ and $\sigma$, tolerance limits are

$$\bar{x} \pm ks,$$

where $k$ is chosen so that we are $100(1-\gamma)\%$ confident that the limits contain at least the proportion $1-\alpha$ of the measurements.

tolerance interval은 population의 majority가 어디에 있는지를 말해준다.

# Example 9.9

Example 9.8에서 $n = 30$, $\bar{x} = 96.2\%$, $s = 0.8\%$이다. Distribution의 central 90%를 포함하는 two-sided 95% tolerance bounds를 구하시오.

From the tolerance table,

$$k = 2.14.$$

Thus,

$$\bar{x} \pm ks = 96.2 \pm (2.14)(0.8).$$

$$94.5 < x < 97.9.$$

우리는 이 구간이 95% confidence로 95% lean beef package distribution의 central 90%를 포함한다고 말할 수 있다.

# Distinction among Three Interval Types

- **Confidence interval**: population mean $\mu$에 대한 구간.
- **Prediction interval**: future single observation에 대한 구간.
- **Tolerance interval**: population의 지정된 비율을 포함하는 구간.

세 구간은 계산식이 비슷해 보여도 해석이 완전히 다르다. 평균이 궁금한지, 다음 관측값이 궁금한지, 모집단 대부분의 위치가 궁금한지 먼저 구분해야 한다.

# Case Study 9.1

A machine produces cylindrical metal pieces. Sample diameters are

$$1.01, 0.97, 1.03, 1.04, 0.99, 0.98, 0.99, 1.01, 1.03.$$

Assume approximately normal distribution. Given

$$\bar{x} = 1.0056, \qquad s = 0.0246.$$

Compute:

1. 99% confidence interval for mean diameter.
2. 99% prediction interval for a single future diameter.
3. 99% tolerance limits containing 95% of produced pieces.

# Case Study 9.1: Solution (1)

For $n = 9$, degrees of freedom $v = 8$.

(a) 99% confidence interval for $\mu$:

$$\bar{x} \pm t_{0.005}\frac{s}{\sqrt{n}} = 1.0056 \pm (3.355)\frac{0.0246}{3} = 1.0056 \pm 0.0275.$$

$$0.9781 < \mu < 1.0331.$$

(b) 99% prediction interval:

$$\bar{x} \pm t_{0.005} s\sqrt{1+\frac{1}{n}} = 1.0056 \pm (3.355)(0.0246)\sqrt{1+\frac{1}{9}}.$$

$$0.9186 < x_0 < 1.0926.$$

# Case Study 9.1: Solution (2)

(c) 99% tolerance limits that contain 95% of the distribution: From the tolerance table,

$$k = 4.550.$$

Thus,

$$\bar{x} \pm ks = 1.0056 \pm (4.550)(0.0246).$$

$$0.8937 < x < 1.1175.$$

해석: confidence interval은 mean diameter에 대한 구간이고, prediction interval은 다음 하나의 metal piece diameter에 대한 구간이며, tolerance interval은 생산되는 metal piece diameter의 central 95%를 포함하는 구간이다.

# 9.8 Two Samples: Estimating the Difference between Two Means

# Estimating $\mu_1 - \mu_2$

For two populations with means $\mu_1, \mu_2$, a point estimator of $\mu_1 - \mu_2$ is

$$\bar{X}_1 - \bar{X}_2.$$

Point estimate:

$$\bar{x}_1 - \bar{x}_2.$$

The sampling distribution of $\bar{X}_1 - \bar{X}_2$ is approximately normal when the sample sizes are reasonable or the populations are normal.

# Confidence Interval for $\mu_1 - \mu_2$: Known Variances

If $\sigma_1^2$ and $\sigma_2^2$ are known,

$$(\bar{x}_1 - \bar{x}_2) - z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}.$$

두 평균의 차이를 추정할 때 standard error는 두 sample mean의 variance를 더해서 계산한다.

# Experimental Conditions and Experimental Unit

When comparing two means, experimental conditions matter.

The **experimental unit** is the part of the experiment that produces experimental error and is responsible for population variance.

Examples:

- drug study: patient or subject
- agricultural experiment: plot of ground
- chemical experiment: quantity of raw materials

조건을 experimental units에 random assignment하면 bias를 줄일 수 있다.

# Example 9.10

Engine A와 Engine B의 gas mileage를 비교하였다. Engine A는 50회, Engine B는 75회 실험하였다. Average gas mileage는 A가 36, B가 42 miles per gallon이었다. Population standard deviations는 A가 6, B가 8이다.

$\mu_B - \mu_A$에 대한 96% confidence interval을 구하시오.

# Example 9.10: Solution

Point estimate:

$$\bar{x}_B - \bar{x}_A = 42 - 36 = 6.$$

For 96%, $\alpha = 0.04$, so $z_{0.02} = 2.05$.

$$6 \pm 2.05\sqrt{\frac{64}{75} + \frac{36}{50}}.$$

$$3.43 < \mu_B - \mu_A < 8.57.$$

두 engine의 평균 gas mileage 차이는 96% confidence로 3.43에서 8.57 miles per gallon 사이로 추정된다.

# Unknown but Equal Variances: Pooled Estimate

When $\sigma_1^2 = \sigma_2^2 = \sigma^2$ but both are unknown, use the pooled estimate.

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}.$$

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{1/n_1 + 1/n_2}}$$

has a t-distribution with

$$v = n_1 + n_2 - 2.$$

# Confidence Interval: Equal Unknown Variances

If the populations are approximately normal with unknown but equal variances, then

$$(\bar{x}_1 - \bar{x}_2) - t_{\alpha/2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{\alpha/2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}.$$

equal variance assumption이 있으면 두 sample variance를 degrees of freedom으로 가중평균하여 pooled variance를 만든다.

# Example 9.11

Cane Creek, Alabama에서 acid mine pollution과 macroinvertebrate species diversity index의 관계를 조사하였다. Downstream station에서 12개 sample의 mean은 3.11, standard deviation은 0.771이었다. Upstream station에서 10개 sample의 mean은 2.04, standard deviation은 0.448이었다.

두 population이 approximately normal이고 equal variances를 가진다고 가정할 때, $\mu_1 - \mu_2$에 대한 90% confidence interval을 구하시오.

# Example 9.11: Solution (1)

Point estimate:

$$\bar{x}_1 - \bar{x}_2 = 3.11 - 2.04 = 1.07.$$

Pooled variance:

$$s_p^2 = \frac{(11)(0.771^2) + (9)(0.448^2)}{12 + 10 - 2} = 0.417.$$

Thus,

$$s_p = 0.646.$$

Degrees of freedom:

$$v = 12 + 10 - 2 = 20.$$

For 90%, $t_{0.05} = 1.725$.

# Example 9.11: Solution (2)

Confidence interval:

$$1.07 \pm (1.725)(0.646)\sqrt{\frac{1}{12} + \frac{1}{10}}.$$

$$0.593 < \mu_1 - \mu_2 < 1.547.$$

두 confidence limit이 모두 positive이므로 downstream station의 average species diversity index가 upstream station보다 크다고 해석할 수 있다.

# Unknown and Unequal Variances

When the population variances are unknown and not assumed equal, use

$$T' = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}.$$

The approximate degrees of freedom are given by the Satterthwaite approximation:

$$\boxed{v = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{(s_1^2/n_1)^2/(n_1 - 1) + (s_2^2/n_2)^2/(n_2 - 1)}.}$$

Round $v$ down to the nearest whole number.

# Confidence Interval: Unequal Unknown Variances

$$(\bar{x}_1 - \bar{x}_2) - t_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}.$$

unequal variance case에서는 pooled variance를 쓰지 않고, 각 sample variance를 따로 사용한다.

# Example 9.12

James River의 두 station에서 orthophosphorus content를 측정하였다. Station 1은 $n_1 = 15$, $\bar{x}_1 = 3.84$, $s_1 = 3.07$이고, Station 2는 $n_2 = 12$, $\bar{x}_2 = 1.49$, $s_2 = 0.80$이다.

Normal populations with different variances를 가정하고, true average orthophosphorus contents의 차이에 대한 95% confidence interval을 구하시오.

# Example 9.12: Solution

Degrees of freedom:

$$v = \frac{(3.07^2/15 + 0.80^2/12)^2}{(3.07^2/15)^2/14 + (0.80^2/12)^2/11} = 16.3 \approx 16.$$

Point estimate:

$$\bar{x}_1 - \bar{x}_2 = 3.84 - 1.49 = 2.35.$$

For 95%, $t_{0.025} = 2.120$ with $v = 16$.

$$2.35 \pm 2.120\sqrt{\frac{3.07^2}{15} + \frac{0.80^2}{12}}.$$

$$0.60 < \mu_1 - \mu_2 < 4.10.$$

우리는 두 station의 true average orthophosphorus contents 차이가 0.60에서 4.10 사이에 있다고 95% confident 할 수 있다.

# 9.9 Paired Observations

# 9.9 Paired Observations

In paired observations, each experimental unit receives both conditions.

Examples:

- weights before and after a diet
- two treatments applied to matched subjects
- two measurements taken on the same object

For each pair, define

$$D_i = X_{1i} - X_{2i}.$$

Then the problem becomes a one-sample problem on the differences.

# Why Pairing Can Help

For paired observations,

$$D_i = X_{1i} - X_{2i}.$$

The variance is

$$\mathrm{Var}(D_i) = \sigma_1^2 + \sigma_2^2 - 2\mathrm{Cov}(X_{1i}, X_{2i}).$$

If observations within a pair are positively correlated, pairing reduces the variance of differences.

pairing의 목적은 experimental unit 간 차이를 제거하여 비교의 precision을 높이는 것이다.

# Confidence Interval for Paired Observations

Let $\bar{d}$ and $s_d$ be the sample mean and standard deviation of the differences.

A $100(1-\alpha)\%$ confidence interval for

$$\mu_D = \mu_1 - \mu_2$$

is

$$\bar{d} - t_{\alpha/2}\frac{s_d}{\sqrt{n}} < \mu_D < \bar{d} + t_{\alpha/2}\frac{s_d}{\sqrt{n}},$$

where $t_{\alpha/2}$ has $v = n - 1$ degrees of freedom.

# Tradeoff in Pairing

Pairing reduces variance, but it also changes the problem into a one-sample problem with $n-1$ degrees of freedom.

Pairing is useful when:

- experimental units are homogeneous within pairs;
- variability between pairs is large;
- the covariance within pair is positive and large.

Pairing may be counterproductive if it reduces variance only slightly while losing degrees of freedom.

# Example 9.13

20명의 Massachusetts Vietnam veterans에 대해 plasma와 fat tissue에서 TCDD level을 측정하였다. 각 veteran에 대해 두 측정값이 있으므로 paired observations이다.

$\mu_1$은 plasma의 true mean TCDD level, $\mu_2$는 fat tissue의 true mean TCDD level이다. Differences가 approximately normal이라고 가정하고 $\mu_1 - \mu_2$에 대한 95% confidence interval을 구하시오.

# Example 9.13: Solution

The differences are defined by

$$d_i = \text{plasma level} - \text{fat tissue level}.$$

From the data,

$$\bar{d} = -0.8700, \qquad s_d = 2.9773, \qquad n = 20.$$

For $v = 19$, $t_{0.025} = 2.093$.

$$-0.8700 \pm (2.093)\frac{2.9773}{\sqrt{20}}.$$

$$-2.2634 < \mu_D < 0.5234.$$

이 interval은 0을 포함하므로 plasma와 fat tissue의 mean TCDD level에 유의한 차이가 있다고 보기 어렵다.

# 9.10 Single Sample: Estimating a Proportion

# 9.10 Estimating a Single Proportion

For a binomial experiment, the point estimator of $p$ is

$$\hat{P} = \frac{X}{n}.$$

The observed sample proportion is

$$\hat{p} = \frac{x}{n}.$$

For large $n$,

$$\hat{P} \approx N\left(p, \frac{pq}{n}\right).$$

sample proportion은 0과 1로 이루어진 Bernoulli data의 sample mean이다.

# Large-Sample Confidence Interval for $p$: Method 1

If $\hat{p} = x/n$ and $\hat{q} = 1 - \hat{p}$, then

$$\hat{p} - z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}} < p < \hat{p} + z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}.$$

Condition for reliability:

$$n\hat{p} \geq 5, \qquad n\hat{q} \geq 5.$$

# Large-Sample Confidence Interval for $p$: Method 2

A more accurate but more complicated form is

$$\frac{\hat{p} + \frac{z_{\alpha/2}^2}{2n}}{1 + \frac{z_{\alpha/2}^2}{n}} - \frac{z_{\alpha/2}}{1 + \frac{z_{\alpha/2}^2}{n}} \sqrt{\frac{\hat{p}\hat{q}}{n} + \frac{z_{\alpha/2}^2}{4n^2}} < p < \frac{\hat{p} + \frac{z_{\alpha/2}^2}{2n}}{1 + \frac{z_{\alpha/2}^2}{n}} + \frac{z_{\alpha/2}}{1 + \frac{z_{\alpha/2}^2}{n}} \sqrt{\frac{\hat{p}\hat{q}}{n} + \frac{z_{\alpha/2}^2}{4n^2}}.$$

$n$이 충분히 크면 Method 1과 Method 2의 차이는 작아진다.

# Example 9.14

Hamilton, Canada에서 TV를 소유한 500가구를 random sample로 조사하였다. 그 중 340가구가 HBO를 구독하고 있었다.

이 도시에서 TV를 소유한 가구 중 HBO를 구독하는 실제 비율 $p$에 대한 95% confidence interval을 구하시오.

# Example 9.14: Solution

Point estimate:

$$\hat{p} = \frac{340}{500} = 0.68, \qquad \hat{q} = 0.32.$$

For 95%, $z_{0.025} = 1.96$.

$$0.68 \pm 1.96\sqrt{\frac{(0.68)(0.32)}{500}}.$$

Using Method 1,

$$0.6391 < p < 0.7209.$$

Using Method 2,

$$0.6378 < p < 0.7194.$$

# Error Bound for Estimating $p$

If $\hat{p}$ is used as an estimate of $p$, then the error is $|\hat{p} - p|$.

We can be $100(1-\alpha)\%$ confident that the error will not exceed

$$z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}.$$

Example 9.14에서는 $\hat{p} = 0.68$이 true proportion $p$와 약 0.04 이하의 차이를 가진다고 95% confident할 수 있다.

# Sample Size for Estimating $p$

If the desired error bound is $e$, then approximately

$$n = \frac{z_{\alpha/2}^2 \hat{p}\hat{q}}{e^2}.$$

If no estimate of $p$ is available, use the maximum possible value $\hat{p}\hat{q} \leq 1/4$.

$$n = \frac{z_{\alpha/2}^2}{4e^2}.$$

# Example 9.15

Example 9.14에서 estimate of $p$가 true value와 0.02 이내가 되도록 95% confident하고 싶다. Preliminary estimate $\hat{p} = 0.68$을 사용하면 sample size는 얼마인가?

$$n = \frac{(1.96)^2(0.68)(0.32)}{(0.02)^2} = 2089.8.$$

따라서

$$n = 2090$$

이 필요하다.

# Example 9.16

Example 9.14에서 preliminary estimate of $p$가 없다고 하자. Estimate가 true value와 0.02 이내가 되도록 적어도 95% confident하려면 sample size는 얼마인가?

Use the conservative formula:

$$n = \frac{(1.96)^2}{4(0.02)^2} = 2401.$$

따라서 preliminary information이 없으면 더 큰 sample size가 필요하다.

# 9.11 Two Samples: Estimating the Difference between Two Proportions

# Estimating $p_1 - p_2$

For two binomial populations, use the difference of sample proportions.

Point estimator:

$$\hat{P}_1 - \hat{P}_2.$$

Point estimate:

$$\hat{p}_1 - \hat{p}_2.$$

For large samples,

$$\hat{P}_1 - \hat{P}_2 \approx N\left(p_1 - p_2, \frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}\right).$$

# Confidence Interval for $p_1 - p_2$

An approximate $100(1-\alpha)\%$ confidence interval is

$$(\hat{p}_1 - \hat{p}_2) - z_{\alpha/2}\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}} < p_1 - p_2 < (\hat{p}_1 - \hat{p}_2) + z_{\alpha/2}\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}.$$

두 비율의 차이도 point estimate ± critical value × standard error의 구조를 가진다.

# Example 9.17

Component part manufacturing process에서 기존 공정과 새로운 공정을 비교한다. Existing process에서는 1500개 중 75개가 defective였고, new process에서는 2000개 중 80개가 defective였다.

Existing process와 new process의 true defective proportion 차이 $p_1 - p_2$에 대한 90% confidence interval을 구하시오.

# Example 9.17: Solution

$$\hat{p}_1 = \frac{75}{1500} = 0.05, \qquad \hat{p}_2 = \frac{80}{2000} = 0.04.$$

Point estimate:

$$\hat{p}_1 - \hat{p}_2 = 0.01.$$

For 90%, $z_{0.05} = 1.645$.

$$1.645\sqrt{\frac{(0.05)(0.95)}{1500} + \frac{(0.04)(0.96)}{2000}} = 0.0117.$$

$$-0.0017 < p_1 - p_2 < 0.0217.$$

The interval contains 0, so there is no clear evidence that the new process significantly decreases the defective proportion.

# 9.12 Single Sample: Estimating the Variance

# 9.12 Estimating a Single Variance

For a normal population, the sample variance $S^2$ is used as a point estimator of $\sigma^2$.

The statistic

$$X^2 = \frac{(n-1)S^2}{\sigma^2}$$

has a chi-squared distribution with

$$v = n - 1$$

degrees of freedom.

variance에 대한 interval은 normal 또는 t가 아니라 chi-squared distribution을 사용한다.

# Confidence Interval for $\sigma^2$

If $s^2$ is the variance of a random sample of size $n$ from a normal population, then

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}.$$

Here the chi-squared values have $v = n - 1$ degrees of freedom and are right-tail values.

A confidence interval for $\sigma$ is obtained by taking square roots of the endpoints.

# Example 9.18

다음은 한 회사가 배포한 grass seed 10개 package의 weights이다.

$$46.4, 46.1, 45.8, 47.0, 46.1, 45.9, 45.8, 46.9, 45.2, 46.0$$

Normal population을 가정하고, weights의 variance에 대한 95% confidence interval을 구하시오.

# Example 9.18: Solution

From the data,

$$s^2 = 0.286, \qquad n = 10, \qquad v = 9.$$

For 95%, using the chi-squared table:

$$\chi^2_{0.025} = 19.023, \qquad \chi^2_{0.975} = 2.700.$$

$$\frac{(9)(0.286)}{19.023} < \sigma^2 < \frac{(9)(0.286)}{2.700}.$$

$$0.135 < \sigma^2 < 0.953.$$

# 9.13 Two Samples: Estimating the Ratio of Two Variances

# 9.13 Estimating $\sigma_1^2/\sigma_2^2$

A point estimate of the ratio of two population variances is

$$\frac{s_1^2}{s_2^2}.$$

For normal populations, use the statistic

$$F = \frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2}.$$

This follows an F-distribution with

$$v_1 = n_1 - 1, \qquad v_2 = n_2 - 1.$$

# Confidence Interval for $\sigma_1^2/\sigma_2^2$

A $100(1-\alpha)\%$ confidence interval for $\sigma_1^2/\sigma_2^2$ is

$$\frac{s_1^2}{s_2^2}\frac{1}{f_{\alpha/2}(v_1, v_2)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{s_1^2}{s_2^2} f_{\alpha/2}(v_2, v_1).$$

variance ratio interval은 F-distribution을 사용하며, 자유도의 순서가 중요하다.

# Example 9.19

Example 9.12에서 station 1과 station 2의 orthophosphorus contents에 대해 unequal variance를 가정하였다. 이를 justify하기 위해 $\sigma_1^2/\sigma_2^2$와 $\sigma_1/\sigma_2$에 대한 98% confidence intervals를 구하시오.

Given:

$$n_1 = 15, \quad n_2 = 12, \quad s_1 = 3.07, \quad s_2 = 0.80.$$

# Example 9.19: Solution

For 98%, $\alpha = 0.02$. From the F-table,

$$f_{0.01}(14, 11) \approx 4.30, \qquad f_{0.01}(11, 14) \approx 3.87.$$

Thus,

$$\left(\frac{3.07^2}{0.80^2}\right)\frac{1}{4.30} < \frac{\sigma_1^2}{\sigma_2^2} < \left(\frac{3.07^2}{0.80^2}\right)(3.87).$$

$$3.425 < \frac{\sigma_1^2}{\sigma_2^2} < 56.991.$$

Taking square roots,

$$1.851 < \frac{\sigma_1}{\sigma_2} < 7.549.$$

Since the interval does not include 1, the unequal variance assumption is reasonable.

# 9.14 Maximum Likelihood Estimation

# 9.14 Maximum Likelihood Estimation

Maximum likelihood estimation chooses the parameter value that makes the observed sample most likely.

For independent observations $x_1, x_2, \ldots, x_n$ from $f(x;\theta)$, the likelihood function is

$$L(x_1, x_2, \ldots, x_n; \theta) = \prod_{i=1}^{n} f(x_i; \theta).$$

maximum likelihood estimator는 관측된 sample이 나올 가능성을 가장 크게 만드는 parameter value이다.

# Likelihood Function: Interpretation

For a discrete distribution, the likelihood can be interpreted as a joint probability.

$$L(x_1, \ldots, x_n; \theta) = P(X_1 = x_1, \ldots, X_n = x_n \mid \theta).$$

For continuous distributions, likelihood is not a probability, but the same maximization idea applies.

Often it is easier to maximize the log-likelihood:

$$\ell(\theta) = \ln L(\theta).$$

# Example 9.20

Poisson distribution의 probability mass function이

$$f(x \mid \mu) = \frac{e^{-\mu}\mu^x}{x!}, \qquad x = 0, 1, 2, \ldots$$

라고 하자. Random sample $x_1, x_2, \ldots, x_n$이 주어졌을 때, $\mu$의 maximum likelihood estimate를 구하시오.

# Example 9.20: Solution

Likelihood:

$$L(x_1, \ldots, x_n; \mu) = \prod_{i=1}^{n} \frac{e^{-\mu}\mu^{x_i}}{x_i!} = \frac{e^{-n\mu}\mu^{\sum_{i=1}^{n} x_i}}{\prod_{i=1}^{n} x_i!}.$$

Log-likelihood:

$$\ln L = -n\mu + \left(\sum_{i=1}^{n} x_i\right) \ln \mu - \ln\left(\prod_{i=1}^{n} x_i!\right).$$

Differentiate:

$$\frac{\partial \ln L}{\partial \mu} = -n + \frac{\sum_{i=1}^{n} x_i}{\mu}.$$

Set equal to zero:

$$\boxed{\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}.}$$

# Example 9.21

Random sample $x_1, x_2, \ldots, x_n$이 normal distribution $N(\mu, \sigma)$에서 나왔다고 하자.

$\mu$와 $\sigma^2$의 maximum likelihood estimators를 구하시오.

# Example 9.21: Solution (1)

Likelihood:

$$L(x_1, \dots, x_n; \mu, \sigma^2) = \frac{1}{(2\pi)^{n/2}(\sigma^2)^{n/2}} \exp\left[-\frac{1}{2}\sum_{i=1}^{n}\left(\frac{x_i - \mu}{\sigma}\right)^2\right].$$

Log-likelihood:

$$\ln L = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln\sigma^2 - \frac{1}{2}\sum_{i=1}^{n}\left(\frac{x_i - \mu}{\sigma}\right)^2.$$

Partial derivative with respect to $\mu$:

$$\frac{\partial \ln L}{\partial \mu} = \sum_{i=1}^{n}\frac{x_i - \mu}{\sigma^2}.$$

Setting this equal to zero gives

$$\hat{\mu} = \bar{x}.$$

# Example 9.21: Solution (2)

Partial derivative with respect to $\sigma^2$:

$$\frac{\partial \ln L}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^{n} (x_i - \mu)^2.$$

Setting this equal to zero gives

$$n\sigma^2 = \sum_{i=1}^{n} (x_i - \mu)^2.$$

Using $\hat{\mu} = \bar{x}$,

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2.$$

주의: MLE of $\sigma^2$ uses denominator $n$, whereas the unbiased sample variance uses denominator $n - 1$.

# Example 9.22

10마리 rats가 biomedical study에 사용되었다. Survival times, in months, are

$$14, 17, 27, 18, 12, 8, 22, 13, 19, 12.$$

Exponential distribution이 적용된다고 가정한다. Mean survival time의 maximum likelihood estimate를 구하시오.

# Example 9.22: Solution

For exponential distribution,

$$f(x;\beta) = \frac{1}{\beta}e^{-x/\beta}, \qquad x > 0.$$

Log-likelihood for $n = 10$:

$$\ln L = -10\ln\beta - \frac{1}{\beta}\sum_{i=1}^{10} x_i.$$

Derivative:

$$\frac{\partial \ln L}{\partial \beta} = -\frac{10}{\beta} + \frac{1}{\beta^2}\sum_{i=1}^{10} x_i.$$

Set equal to zero:

$$\hat{\beta} = \bar{x} = 16.2.$$

따라서 mean survival time의 MLE는 16.2 months이다.

# Example 9.23

Sample values

$$12, 11.2, 13.5, 12.3, 13.8, 11.9$$

come from the density

$$f(x;\theta) = \frac{\theta}{x^{\theta+1}}, \qquad x > 1,$$

where $\theta > 0$. Maximum likelihood estimate of $\theta$를 구하시오.

# Example 9.23: Solution

Likelihood:

$$L(x_1, \ldots, x_n; \theta) = \prod_{i=1}^{n} \frac{\theta}{x_i^{\theta+1}} = \frac{\theta^n}{\left(\prod_{i=1}^{n} x_i\right)^{\theta+1}}.$$

Log-likelihood:

$$\ln L = n \ln \theta - (\theta + 1) \sum_{i=1}^{n} \ln x_i.$$

Derivative:

$$\frac{\partial \ln L}{\partial \theta} = \frac{n}{\theta} - \sum_{i=1}^{n} \ln x_i.$$

Set equal to zero:

$$\hat{\theta} = \frac{n}{\sum_{i=1}^{n} \ln x_i} = \frac{6}{\ln(12) + \ln(11.2) + \ln(13.5) + \ln(12.3) + \ln(13.8) + \ln(11.9)} = 0.3970.$$

# Additional Comments on Maximum Likelihood

Maximum likelihood estimation requires knowledge of the underlying distribution.

- MLEs are not necessarily unbiased.
- MLEs often have desirable large-sample properties.
- MLEs are widely used because the method applies broadly once the model is specified.

MLE는 관측된 데이터를 가장 그럴듯하게 만드는 parameter를 선택하는 방법이지만, unbiasedness가 자동으로 보장되지는 않는다.

# 9.15 Potential Misconceptions and Hazards

# 9.15 Potential Misconceptions and Hazards (1)

A large-sample confidence interval on $\mu$ is often misused.

The formula

$$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$$

is based on the Central Limit Theorem and the approximation $s \approx \sigma$.

- It should not be used blindly for small samples.
- $n$ should generally be at least 30.
- The underlying distribution should not be extremely skewed.

# 9.15 Potential Misconceptions and Hazards (2)

The t-distribution is used when $\sigma$ is unknown, but it assumes that the sampled population is normal.

The t-procedure is reasonably robust to moderate deviations from normality.

However:

- severe nonnormality can be problematic;
- normal probability plots or goodness-of-fit tests may be needed.

t-interval은 어느 정도 robust하지만, 극단적인 nonnormal distribution에는 조심해서 사용해야 한다.

# 9.15 Potential Misconceptions and Hazards (3)

One serious misuse is confusing different interval interpretations.

- Confidence interval: about the parameter, usually a mean.
- Prediction interval: about one future observation.
- Tolerance interval: about a specified proportion of the population.

문제에서 알고 싶은 대상이 mean인지, next observation인지, population의 majority인지 먼저 판단해야 한다.

# 9.15 Potential Misconceptions and Hazards (4)

A confidence interval should not be interpreted as "the probability that the fixed parameter lies in this computed interval."

Frequentist interpretation: If the same procedure is repeated many times, approximately $100(1-\alpha)\%$ of the resulting intervals will contain the true parameter.

또 다른 주의점: chi-squared interval for variance is not robust to nonnormality. Normality should be checked carefully when using variance intervals.